🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 7 · STANDARD 6.NOS.9**

# Represent Polygons on the Coordinate Plane

Lesson 7-7 · Teacher Copy

**Name:** Type your name **Date:** Today's date **Class:** Class period

## My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.**
**Language Objective: I can explain how I found a distance using the words vertices, coordinates, and absolute value.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

vertex | polygon | perimeter | distance between vertices | scale

👆 *Tap any word to see what it means and a picture.*

*Level 3: try the blanks from memory first, then check the bank.*

1. __________ — A corner point of a polygon, named by an ordered pair on the coordinate plane.

2. __________ — A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.

3. __________ — The total distance around a polygon — the sum of all its side lengths.

4. __________ — When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.

**5** ______ — What one grid unit stands for in the real situation.

## Write About the Math

**Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

**EXPLAIN**

**Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- [ ] **vertex** — A corner point of a polygon, named by an ordered pair on the coordinate plane.
  *vértice — Un punto de esquina de un polígono, nombrado por un par ordenado en el plano de coordenadas.*
- [ ] **polygon** — A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.
  *polígono — Una figura cerrada hecha de segmentos; en el plano se dibuja marcando los vértices y conectándolos en orden.*
- [ ] **perimeter** — The total distance around a polygon — the sum of all its side lengths.
  *perímetro — La distancia total alrededor de un polígono: la suma de todos sus lados.*
- [ ] **distance between vertices** — When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.
  *distancia entre vértices — Cuando dos vértices comparten una coordenada x o y, la distancia es la diferencia absoluta de las otras coordenadas.*
- [ ] **scale** — What one grid unit stands for in the real situation.
  *escala — Lo que representa una unidad de la cuadrícula en la situación real.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.
*Di tu idea a un compañero o en voz baja antes de escribir.*

**The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.**

*Di tu respuesta primero: Mi respuesta es ____ porque ____.*

## 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** When two vertices share the same y-coordinate, the segment between them is horizontal, so its length comes from the DIFFERENCE in the coordinate that changes — the x-coordinates.

**Evidence:** A strong answer explains that the SHARED coordinate tells you the direction of the side, and the DIFFERING coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

**Reasoning:** This evidence connects the definition of polygon to the mathematical claim.

### Start

**Most language support**

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.**
- **My answer is ____.**
  *Mi respuesta es ____.*

### Build

**Some language support**

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **I know because ____.**
  *Lo sé porque ____.*
- **So ____.**
  *Entonces ____.*

### Explain

**Light language support**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **My math evidence is ____.**
  *Mi evidencia matemática es ____.*
- **This proves ____ because ____.**
  *Esto demuestra ____ porque ____.*

## 4. Check Your Explanation

- [ ] I answered the question.
  *Respondí la pregunta.*
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  *Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- [ ] I explained how the evidence proves my answer.
  *Expliqué cómo la evidencia demuestra mi respuesta.*
- [ ] I used at least two math words.
  *Usé por lo menos dos palabras matemáticas.*
- [ ] I reread complete sentences and punctuation.
  *Volví a leer mis oraciones completas y la puntuación.*

## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

## Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from $y = 3$ down to $y = -3.75$ the distance is $3 + 3.75 = 6.75$ units, not $3.75 - 3 = 0.75$. Distances that cross zero are the whole reason the absolute value is there.*
7. **Try It 1:**
8. **Try It 2:**